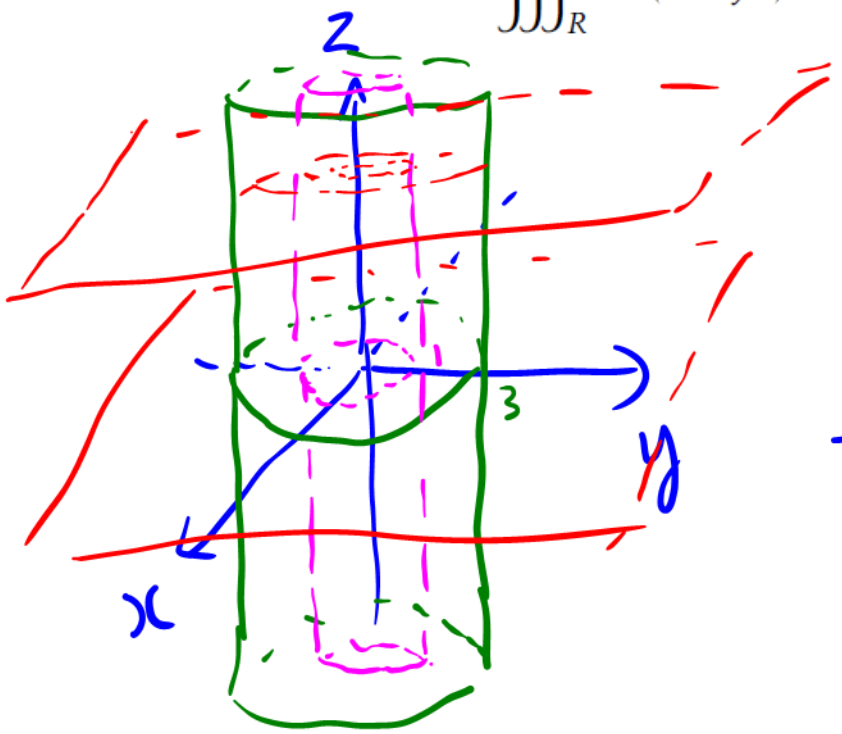
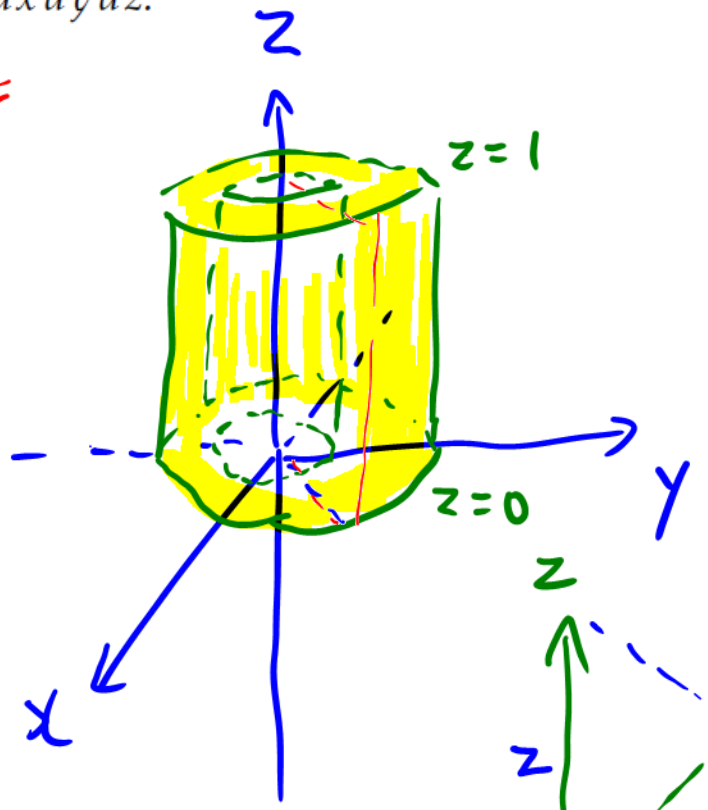
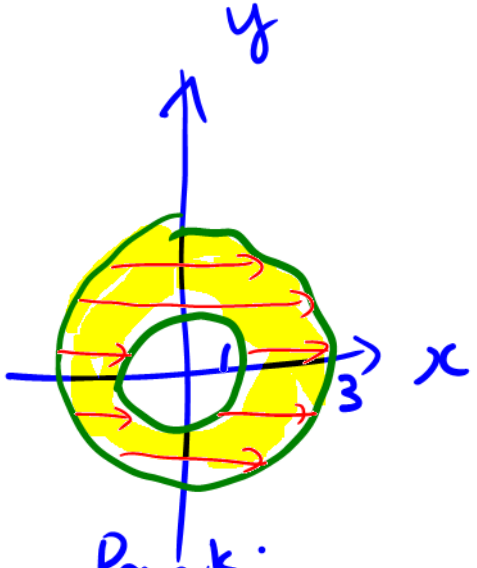
25. Let R be the solid region in the xyz-space bounded by the surface $x^2+y^2 = 9$, the surface $x^2+y^2 = 1$, the plane $z = 1$ and the plane $z = 0$. Compute the integral

$$\iiint_R z^2 (x^2+y^2)^2 \, dx\, dy\, dz.$$

$\frac{728\pi}{9}$



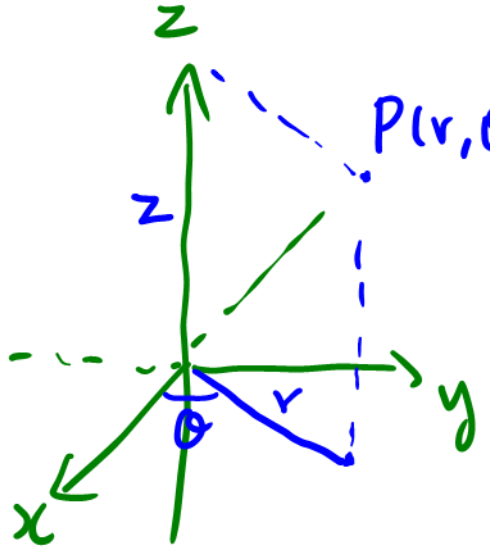




Remark:
Horizontal arrows break into 2 arrows.
∴ difficult to use rectangular coordinates.

cylindrical coordinates (r, θ, z)

$x = r\cos\theta,\ y = r\sin\theta,\ z = z$

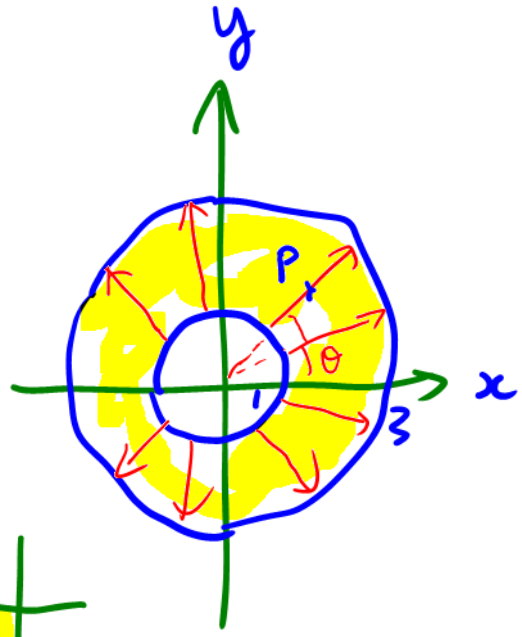


$$dV = dx\,dy\,dz = r\,dz\,dr\,d\theta$$

$$r = \sqrt{x^2+y^2}$$

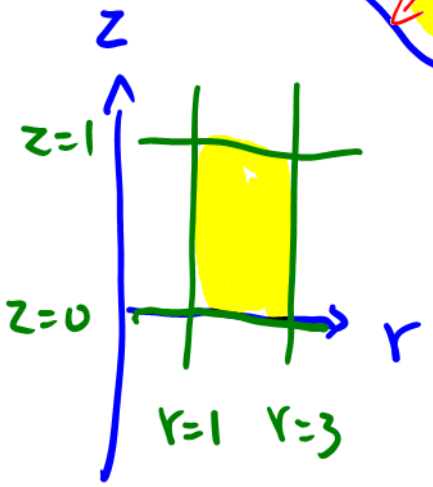
$$R = \{(x,y,z) : 1 \le x^2+y^2 \le 9,\ 0 \le z \le 1\}$$

$$= \{(r,\theta,z) : 0 \le \theta \le 2\pi,\ 1 \le r \le 3,\ 0 \le z \le 1\}$$



$$\iiint_R z^2(x^2+y^2)^2\,dV$$

$$= \int_{\theta=0}^{\theta=2\pi} \int_{r=1}^{r=3} \left(\int_{z=0}^{z=1} z^2 r^4 \cdot r\,dz \right) dr\,d\theta$$



$$= \int_{\theta=0}^{\theta=2\pi} \int_{r=1}^{r=3} r^5 \left[\frac{z^3}{3}\right]_{z=0}^{z=1} dr\,d\theta$$

$$= \left[\frac{z^3}{3}\right]_{z=0}^{z=1} \cdot \int_{\theta=0}^{\theta=2\pi} \int_{r=1}^{r=3} r^5\,dr\,d\theta$$

$$= \left[\frac{z^3}{3}\right]_{z=0}^{z=1} \cdot \int_{\theta=0}^{\theta=2\pi} \left[\frac{r^6}{6}\right]_{r=1}^{r=3} d\theta$$

$$= \left[\frac{z^3}{3}\right]_{z=0}^{z=1} \cdot \left[\frac{r^6}{6}\right]_{r=1}^{r=3} \cdot \left[\theta\right]_{\theta=0}^{\theta=2\pi}$$

$$= \frac{1-0}{3} \cdot \frac{3^6-1}{6} \cdot 2\pi$$

$$= \frac{1}{3} \cdot \frac{728}{6_3} \cdot \cancel{2}\pi$$

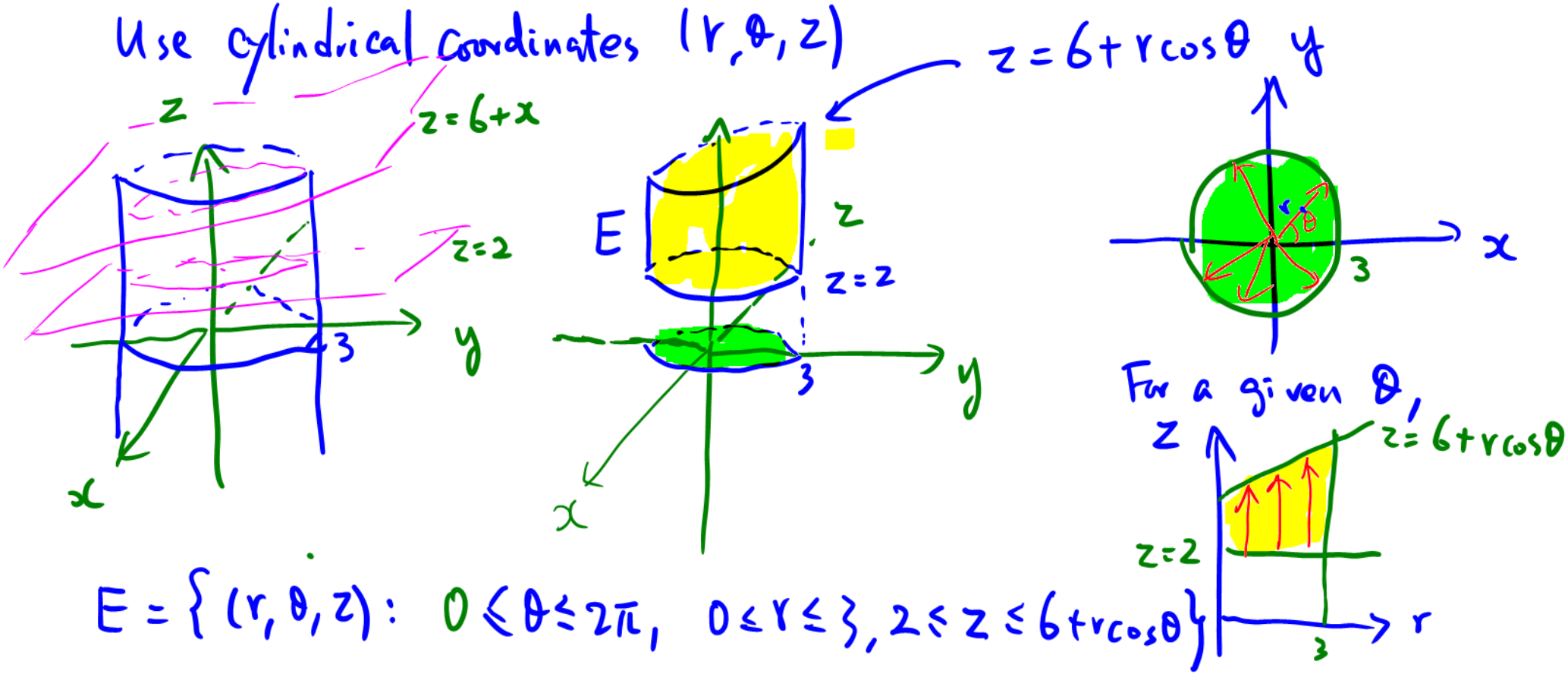
$$= \frac{728\pi}{9}.$$

26. Consider the solid region E in the xyz-space bounded by the cylinder $x^2+y^2 = 9$, the horizontal plane $z = 2$, and the surface $z = 6 + x$. Evaluate the triple integral

$$\frac{1}{\pi}\iiint_E (x^2+y^2)\,dV.$$

Give the answer as an integer or a fraction in lowest terms.

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$E = \{(r,\theta,z):\ 0 \le \theta \le 2\pi,\ 0 \le r \le 3,\ 2 \le z \le 6 + r\cos\theta\}$

z : innermost variable

$$\frac{1}{\pi}\iiint_E (x^2+y^2)\,dV$$

$$= \frac{1}{\pi}\int_{\theta=0}^{2\pi}\int_{r=0}^{r=3}\left(\int_{z=2}^{z=6+r\cos\theta} r^2\cdot r\,dz\right)dr\,d\theta$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=3} r^3\Big[z\Big]_{z=2}^{z=6+r\cos\theta}\,dr\,d\theta$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=3} r^3\left(6+r\cos\theta-2\right)dr\,d\theta$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\left(\int_{r=0}^{r=3}\left(4r^3+r^4\cos\theta\right)dr\right)d\theta$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\left[r^4+\frac{r^5}{5}\cos\theta\right]_{r=0}^{r=3}d\theta$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\left(3^4+\frac{3^5}{5}\cos\theta-(0+0)\right)d\theta$$

$$= \frac{1}{\pi}\left[81\theta + \frac{243}{5}\sin\theta\right]_{\theta=0}^{\theta=2\pi}$$

$$= \frac{1}{\pi}\left[(162\pi + 0) - (0+0)\right]$$

$$= 162.$$

27. Let E be the unit ball consisting of points (x, y, z) in the xyz-space such that $x^2+y^2+z^2 \le 1$. Evaluate the triple integral

$$\iiint_E (x^2+y^2+z^2)\,dV.$$

Give the answer in terms of π.

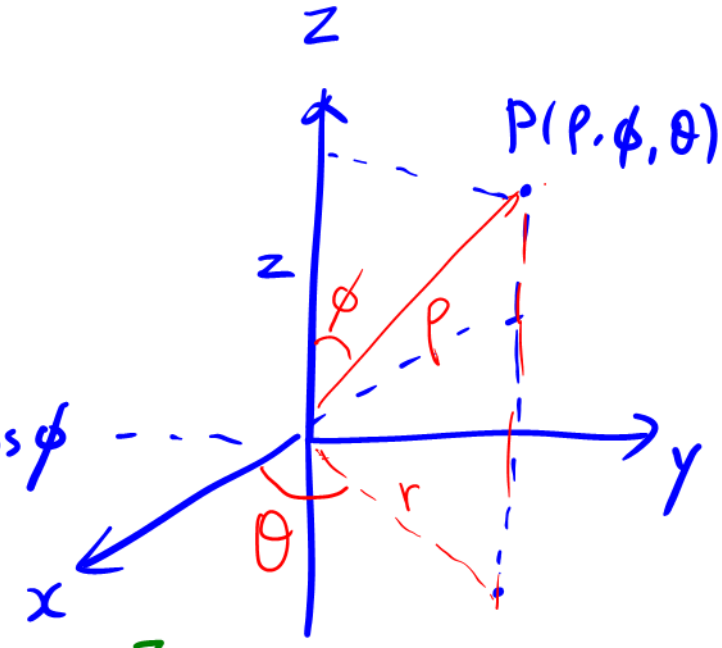
$\frac{4\pi}{5}$

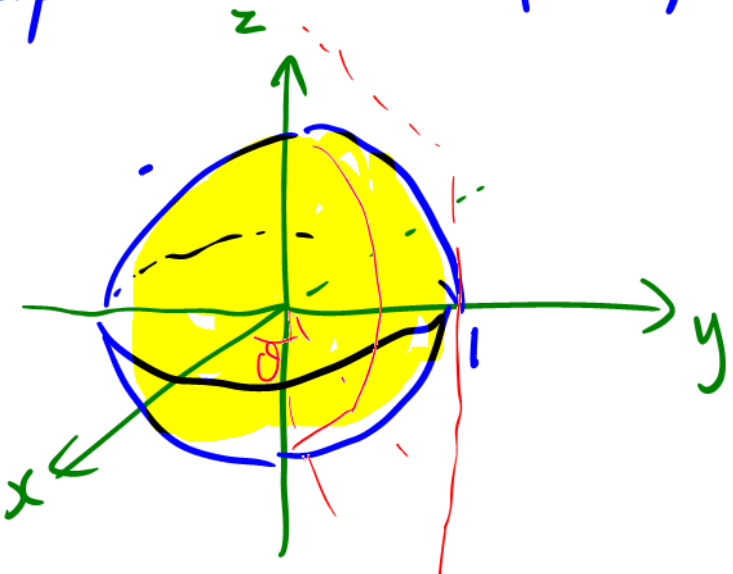
Use spherical coordinates $(\rho, \phi, d\theta)$

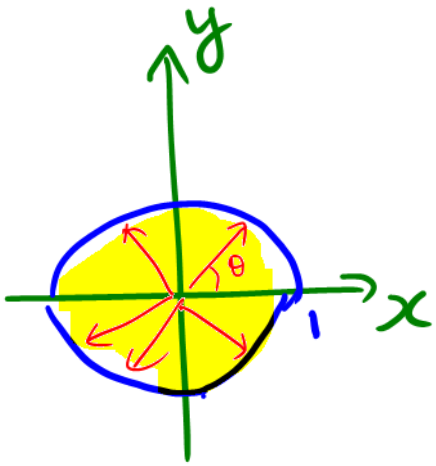
$x = \rho\sin\phi\cos\theta,\ y = \rho\sin\phi\sin\theta,\ z = \rho\cos\phi$

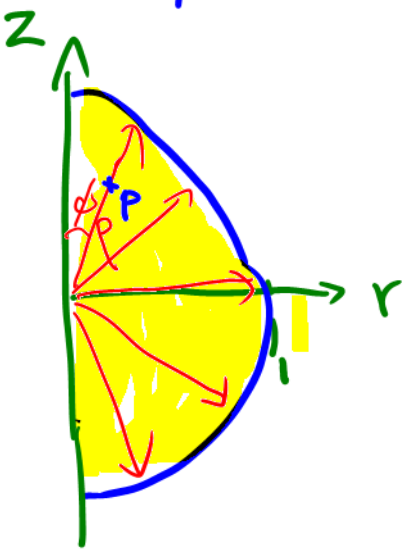
$dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

$r = \rho\sin\phi$









$$E = \{(\rho, \phi, \theta) : 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi,\ 0 \le \rho \le 1\}$$

$$\iiint_E (x^2+y^2+z^2)\, dV$$

$$= \int_{\theta=0}^{\theta=2\pi} \int_{\phi=0}^{\phi=\pi} \int_{\rho=0}^{\rho=1} \rho^2 \cdot \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi} \int_{\phi=0}^{\phi=\pi} \sin\phi \left[\frac{\rho^5}{5}\right]_{\rho=0}^{\rho=1} d\phi \, d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi} \int_{\phi=0}^{\phi=\pi} \sin\phi \cdot \frac{1-0}{5} \, d\phi \, d\theta$$

$$= \frac{1}{5} \cdot \int_{\theta=0}^{\theta=2\pi} \left[-\cos\phi\right]_{\phi=0}^{\phi=\pi} d\theta$$

$$= \frac{1}{5} \cdot \int_{\theta=0}^{\theta=2\pi} \left(1-(-1)\right) d\theta$$

$$= \frac{1}{5} \cdot \left[2\theta\right]_{\theta=0}^{\theta=2\pi} = \frac{4\pi}{5}.$$

28. Let E be the solid region in the xyz-space consisting of points (x, y, z) such that $\sqrt{x^2+y^2} \le z \le 2$. Rewrite the triple integral

$$\iiint_E \frac{z}{(x^2+y^2+z^2+2)^2}\, dV$$

as an iterated integral in spherical coordinates and with integration order $d\rho\, d\phi\, d\theta$.

[You do not have to compute the integral.]

Give the answer in the form

$$\int_a^b \int_c^d \int_e^f g(\rho, \phi, \theta)\, d\rho\, d\phi\, d\theta,$$

where a and b are real numbers, c and d are expressions in terms of θ, e and f are expressions in terms of θ and ϕ, and $g(\rho, \phi, \theta)$ is an expression in terms of ρ, ϕ, θ. Note that some of c, d, e, f may also be real numbers.

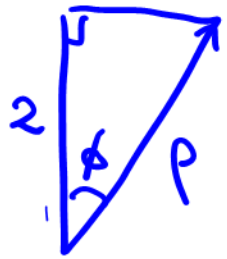
$$\int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^{2\sec\phi} \frac{\rho^3 \cos\phi \sin\phi}{(\rho^2+2)^2}\, d\rho\, d\phi\, d\theta$$

2 bounding surfaces for E :

(i) $\sqrt{x^2+y^2} = z$

$\Leftrightarrow \quad r = z$

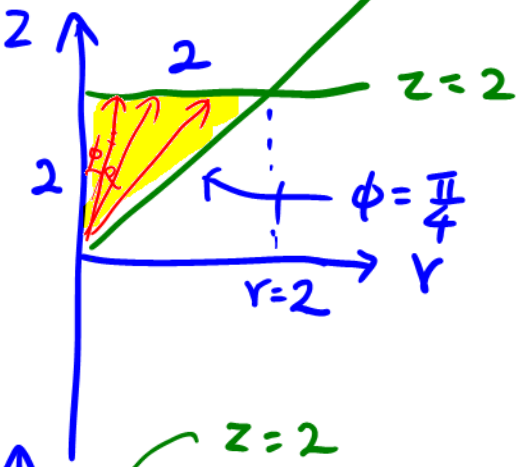
(ii) $z = 2$



$\rho\cos\phi = 2$

$\rho = 2\sec\phi$

For each θ, $z = r$

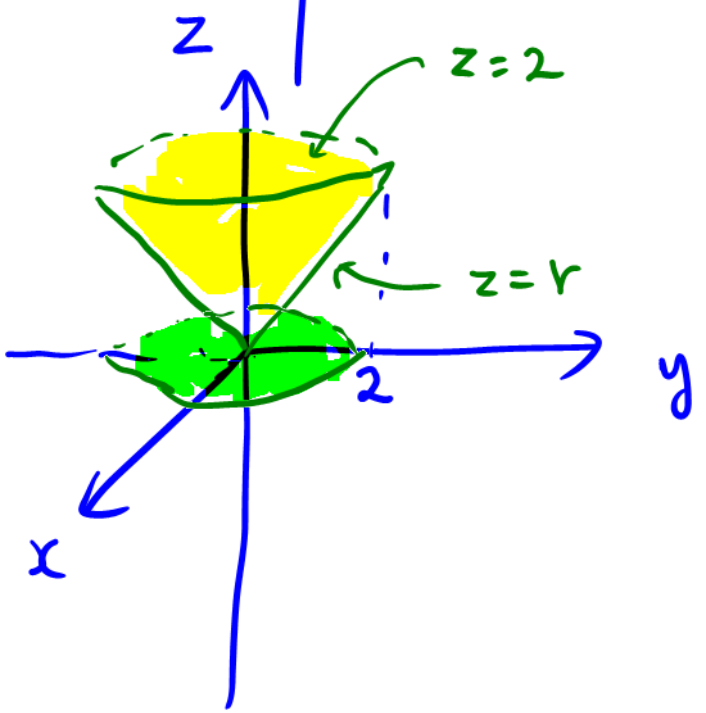


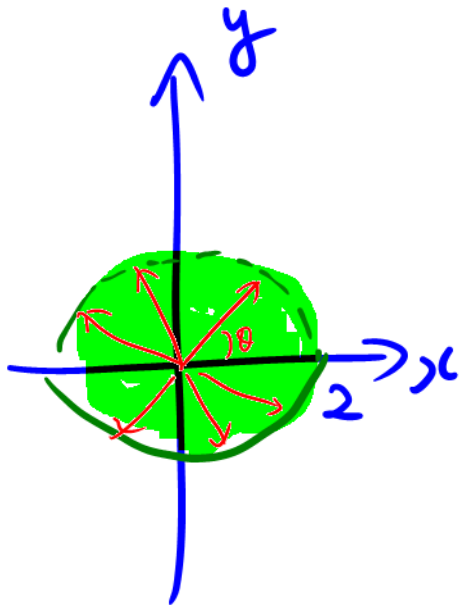
For $\rho > 0$ $z = r$

$\Leftrightarrow \quad \rho\cos\phi = \rho\sin\phi$

$\Leftrightarrow \quad \tan\phi = 1$

$\Leftrightarrow \quad \phi = \frac{\pi}{4}$





$$E = \{(\rho, \phi, \theta) : 0 \le \theta \le 2\pi,\ 0 \le \phi \le \tfrac{\pi}{4},\ 0 \le \rho \le 2\sec\phi\}$$

$$\iiint_E \frac{z}{(x^2+y^2+z^2+2)^2}\,dV$$

$$= \int_{\theta=0}^{\theta=2\pi} \int_{\phi=0}^{\phi=\frac{\pi}{4}} \int_{\rho=0}^{\rho=2\sec\phi} \frac{\rho\cos\phi}{(\rho^2+2)^2} \cdot \rho^2 \sin\phi \; d\rho\, d\phi\, d\theta$$